

Stochastic Modeling of Bridge Deterioration Using Classification Tree and Logistic Regression

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Abstract: This paper presents a new method to develop stochastic deterioration models using a combination of methods including Markov chains, logistic regression, and classification trees. It is computationally more efficient to use logistic regression with the Markov chain process than it is to use optimization-based approaches, and the former is shown to marginally improve the prediction of condition ratings for small data sets. Annually inspected bridge data are split into groups using a classification tree, and logistic regression is used to determine transition probabilities for a Markov chain process. A case study was conducted to determine the effectiveness of using the proposed logistic regression and Markov chain approach for the small data sets created by the classification tree. Wyoming bridge inspection data were split into 15 subsets based on 5 explanatory variables, and deterioration models were developed for each subset. Error analysis showed that logistic regression performed marginally better than traditional methods when estimating the transition probability matrix when limited data are accessible. DOI: [10.1061/\(ASCE\)IS.1943-555X.0000466](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000466). © 2018 American Society of Civil Engineers.

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Introduction

The National Bridge Inspection Program aims to maintain performance of the National Bridge Inventory (NBI) at satisfactory levels for public use and safety. Inspection results are often organized by element-level condition rating and are used to estimate the overall condition of bridge inventory. Bridge deterioration models have been widely adopted throughout the United States in order to quantify deficiency levels and effectively manage bridge inventories. Deterioration models are used to predict generalized condition ratings for bridge components and to support owner or agency decision-making processes. Deterioration predictions assist in establishing maintenance plans that can include repair, replacement, rehabilitation, and lifecycle cost analysis. Numerous studies have noted the importance of accurately predicting the condition rating of deteriorating structures to estimate maintenance costs (Estes and Frangopol 2001; Grall et al. 2002; Ranjith et al. 2011). While the observed improvement is incremental, even small improvements are beneficial.

The deterioration process consists of the combined effects of physical and chemical changes due to reinforcement corrosion, concrete creep and shrinkage, fatigue, and numerous other factors (Abed-Al-Rahim and Johnston 1995). Due to the difficulties of measuring the effects of these complex phenomena, bridge

deterioration models are mostly described as a function of condition ratings versus time, considering explanatory variables such as age, traffic, weather, and functional classification (Ben-Akiva and Gopinath 1995; Ford et al. 2012; Bu et al. 2014). The National Bridge Inspection Standards (NBIS) provide guidelines for the inspection qualifications and intervals, and many studies have adopted the NBIS inspection categories to introduce explanatory variables and condition ratings (Chase and Gáspár 2000; Mauch and Madanat 2001; Hatami and Morcoux 2015; Saeed et al. 2017). Based on the NBIS guidelines, the inspection categories are specified into 116 bridge items and the inspection interval is usually set at 2 years. The element level of inspection results (condition and appraisal ratings) is included in the NBIS, and their ratings are separated into 10 levels. In general, the deck, superstructure, and substructure condition ratings are used to develop deterioration models that give a rough classification, but are sufficiently effective and comprehensive for bridge owners and operating agencies (FHWA 2011).

Bridge deterioration models can be classified as deterministic or stochastic, depending upon how the method used handles uncertainty. Deterministic deterioration models are the best models for describing bridge deterioration and incorporate regression model equations associated with explanatory variables (Veshosky et al. 1994; Hatami and Morcoux 2011; Chang et al. 2017). Deterministic deterioration models are infrequently used for bridge management, but are convenient because they show the effects of explanatory variables. Stochastic models introduce random variables to reflect uncertainty in describing the complex deterioration process. The Markov chain process is one of the most well-known methods used to model deterioration processes and has been adopted by many researchers and management agencies (Jiang et al. 1988; Cesare et al. 1992; Madanat et al. 1995; Frangopol et al. 2004; Baik et al. 2006; Bu et al. 2014). The Markov chain is classified as a state-based approach that estimates the deterioration probability based on the condition rating changes within adjacent inspection years. Time-based approaches have been developed using Weibull functions to estimate the sojourn time of a

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bridge at a specific condition level (Mishalani and Madanat 2002; Sobanjo et al. 2010).

The classification concept is widely used to track the physical performance of structural systems and to provide guidelines for bridge asset management. For in-state bridge assets, a Weibull distribution can be used to quantify the sojourn time for each of the condition states and various bridge categories (Agrawal et al. 2010; Hatami and Morcouc 2011; Sobanjo 2011). Uncertainty contribution of the sojourn time for each condition state can be used to develop deterioration models for various bridge categories. Ford et al. (2012) analyzed bridges in all US states and estimated their life expectancies depending on the element, region, condition ratings, and age groups used. The classification concept was also adopted for various management needs such as wastewater and sewer pipe monitoring (Baik et al. 2006; Koo and Ariaratnam 2006).

The objective of this study was to compare stochastic deterioration models especially suited for small data sets based on a Markov chain. The comparison was performed on Wyoming bridge deck element inspection data that were collected through NBIS. A penalized linear regression technique was implemented to determine the relative importance of candidate variables (Chang et al. 2017). A classification tree was used to split a set of bridges into multiple subsets based on explanatory variables (Lewis 2000; Pittou et al. 2009). For a subset of bridges, the inspection data were grouped by an identical age period in a process called zoning (Butt et al. 1987; Hong et al. 2013). The classification and zoning techniques reduce the size of inspection data and create transition probability matrices that can provide more accuracy when predicting condition rating, but subsets can potentially become too small to represent deterioration properly. Logistic regression was implemented to investigate its effectiveness in modeling statistical uncertainties from small categorical or discrete data sets to develop Markov transition probabilities (Ariaratnam et al. 2001). A Markov chain was used to model the deterioration behavior for each group.

The prediction of condition rating was used to evaluate the performance of the proposed method; the results were compared to the results of traditional methods, including the percentage prediction and the optimization approach. The error comparison between the prediction and the actual inspection of the condition rating shows that better performance is achieved using logistic regression, especially for small data sets.

Data Processing

The Wyoming Department of Transportation has inspected its bridge inventory biannually since 1992. As of the end of 2014, the Wyoming bridge inventory totaled 3,127 (FHWA 2014). Wyoming was one of 15 states in the United States where the rate of deficient bridges increased since 2011 (Davis et al. 2013). Although only 0.51% of the nation's bridges are in Wyoming, the proportion of deficient bridges from Wyoming is reported as 12% and it is ranked as 14th highest in number of deficient bridges among the states.

In order to manage the NBI data, Chang et al. (2017) proposed a framework by which explanatory variables can be selected from candidate variables. The relative importance of candidate variables was estimated using penalized linear regression. The research presented in this paper is focused on data classification and the investigation of stochastic deterioration models for small data sets. In order to develop accurate deterioration models, the bridge inventory was split into multiple subsets according to a hierarchy of explanatory variables by utilizing a classification tree (Lewis 2000). A brief description of penalized linear regression known as least

absolute shrinkage and selection operator (LASSO) and the classification tree approach is provided below.

Penalized Linear Regression Model

A linear regression model used to predict condition ratings, $\mathbf{y}$, for m number of bridges is defined as follows:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (1)$$

In Eq. (1), $\mathbf{X} \in \mathcal{R}^{m \times n}$ denotes bridge observation data for n explanatory variables; $\boldsymbol{\beta} \in \mathcal{R}^n$ is the regression coefficient vector minimizing the mean squared error of $\boldsymbol{\varepsilon} \in \mathcal{R}^n$.

Although the inclusion of a large set of explanatory variables generally minimizes the regression error, it also brings duplicated or even redundant information that compromises statistical and computational efficiency.

In order to determine the significant variables and to establish a ranking system by which to select explanatory variables, a penalized linear regression was implemented. LASSO is one of the most well-known penalized regression techniques that introduces a shrinkage effect when minimizing the least-squared sum of error (Tibshirani 1996). The estimated coefficient vector for LASSO, $\hat{\boldsymbol{\beta}}_L$, is defined as

$$\hat{\boldsymbol{\beta}}_L = \arg \min \left[(\tilde{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta})^T (\tilde{\mathbf{y}} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^n |\beta_j| \right] \quad (2)$$

In Eq. (2), $\lambda \geq 0$ is a tuning parameter that controls the amount of shrinkage, and the absolute sum (known as L_1 norm) of the coefficient vector is called a penalty or regularizer that induces sparsity, i.e., zero estimates (Tibshirani 1996). The amount of shrinkage is a crucial factor in designing a regression model such that the number of selected explanatory variables generally decreases when the value of λ increases.

Chang et al. (2017) performed a LASSO regression on 2014 inspection data and identified a ranking system for significant variables for the deck, superstructure, and substructure elements, which are sensitive to changes in the condition rating. The same analysis was performed on data from other recent years and the same explanatory variables were selected using the methodology. For this reason, the classification tree discussed below is made of *year built (age)*, *type of wearing surface*, *structure length*, *functional classification of inventory route*, *average daily traffic*, and *lanes on the structure*, which are ranked as the top six explanatory variables for the deck elements by penalized regression. Different locations, bridge design philosophies, and maintenance practices will change the important explanatory variables for different agencies: the explanatory variables listed above are specific to the Wyoming data set. Based on the authors' experience, these variables are also identified as significant variables for other states' inventories.

Classification Tree

In statistics, classification is a collection of techniques that identifies a set of categories or subpopulations from a pooled population. If the total population is a mixture of multiple subpopulations that come from different distributions, appropriate classification could significantly improve the accuracy of the subsequent analysis. The four main components of the classification tree are categorical outcome, predictor variables, learning data set, and future data set (Lewis 2000). For bridge deterioration models, those components correspond to the deterioration levels, explanatory variables, bridge inventory, and the condition rating prediction, respectively.

The entire bridge inventory is split into multiple subsets using explanatory variables pertaining to Wyoming bridges. For deck elements, the first explanatory variable is the *type of wearing surface*, which includes 10 types, according to NBI data. Over 85% of the total bridges fall into one of three types: WS3 (latex concrete or similar additive), WS6 (bituminous), and WS0 (none—meaning no additional concrete thickness or wearing surface was included in the deck). Fewer than 50 bridges are of the types monolithic concrete, integral concrete, low slump concrete, and other as of 2014, see Fig. 1(a). The second explanatory variable is *structure length*, of which indices (SL1–SL5) are defined for every 50 m of span length in this research. For example, SL1 includes bridges having main span lengths between 0 and 50 m. The distribution of 2014 NBIS is shown in Fig. 1(b), where the proportion of bridges longer than 100 m is less than 5% of the total; therefore, this classification is meaningful only for SL1 and SL2. The next ranked explanatory variables in sequence are *functional classification of inventory route*, *average daily traffic*, and *lanes on the structure*. The proportional distribution for bridges associated with these explanatory variables is shown in Figs. 1(c–e), respectively. Almost 85% of principal arterial (FC01 and FC02), major collector (FC07), and local (FC09) bridges are located in a rural area. *Average daily traffic* is defined for every 5,000 vehicles per day (ADT1–ADT5), and the distribution illustrates that most Wyoming bridges are carrying less than 10,000 per day.

The logic of a classification tree is illustrated in Fig. 2. The first explanatory variable (*type of wearing surface*) enables the development of six unique deterioration models for [WS3, WS5, WS6, WS7, WS8, WS0] that contain enough data. The other bridges belonging to [WS1, WS2, WS4, WS9] were combined to develop an extra deterioration model with sufficient data. For convenience, this process is called the first level of classification. The next explanatory variable (*structure length*) splits the bridges belonging to [WS3, WS6, WS7, WS8, WS0] into multiple subsets. Accordingly, this is called the second level of classification. Within the five explanatory variables for Wyoming bridges are 15 subsets. For example, the first subset contains 178 bridges and is associated with the explanatory variables WS3 = latex concrete/similar, $0 < SL1 < 50$ m; FC01 = rural (principal arterial—interstate); $0 < ADT1 < 5,000$; and LOS2 = two lanes.

The classification tree can be extended to include more levels depending on the number of bridges and the management policy. In this study, the maximum level is set at the fifth level for practicality. The fourth explanatory variable (*annual daily traffic*) and the fifth explanatory variable (*lanes on the structure*) create almost the same groups as the previous levels, indicating that further classification is trivial for Wyoming bridges due to their low quantity.

Development of Stochastic Deterioration Models

The element level of inspection data is converted into NBI condition rating using the mapping criterion outlined by Hearn et al. (1993). The Wyoming bridge inspection system follows the NBIR rating system and rates the condition of bridge elements on a scale from 1 (imminent failure) to 9 (excellent). The condition ratings gradually decrease as the age increases unless a maintenance action such as repair, replacement, or rehabilitation is applied. The Markov chain process is the most common technique utilized to generalize the decrease in condition ratings over time and is extensively used for many applications (Agrawal et al. 2010; Bocchini et al. 2013).

In order to consider the *year built* (*age*) variable in the deterioration model, the zoning technique was implemented to split the

bridges into subsets for each decade. Markov chain processes were used for each zone where logistic regression was used to determine the transition probabilities. Fig. 3 illustrates the step-by-step framework used to develop the stochastic deterioration models in this study.

Markov Chain-Based Approach

A typical Markov chain for stationary bridge deterioration considers nine discrete states, and transitions occur between any two states (Sinha et al. 1988; Cesare et al. 1992). The transition state matrix is defined as

$$\mathbf{P} = \begin{bmatrix} p_{11} & 1-p_{11} & 0 & \cdots & 0 \\ 0 & p_{22} & 1-p_{22} & \cdots & 0 \\ 0 & 0 & p_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \quad (3)$$

In Eq. (3), p_{ii} denotes the probability of an element remaining at state i in one discretized time step (annual event for bridge maintenance). Using the total probability theory, the n th stage state probability, $\mathbf{q}_n$, is calculated as

$$\mathbf{q}_n = \mathbf{q}_{n-1}\mathbf{P} = \mathbf{q}_{n-2}\mathbf{P}^2 = \cdots = \mathbf{q}_0\mathbf{P}^n \quad (4)$$

In Eq. (4), $\mathbf{q}_0$ is the state probability vector at the initial stage. Defining $\mathbf{R}$ as $[9 \ 8 \ 7 \ \cdots \ 1]$, the expectation of a condition rating at the n th stage is defined as

$$\mathbf{R}_{P,n} = \mathbf{q}_n \times \mathbf{R}^T \quad (5)$$

Traditionally, the elements of a transition probability matrix can be obtained by either percentage prediction or optimization methods (Butt et al. 1987; Jiang et al. 1988). For percentage prediction, let $m_{i,j}$ be the count for the transition change from condition state i to j . The element of the transition probability matrix $p_{i,j}$ is defined as

$$p_{ij} = m_{i,j} / \sum_j m_{i,j} \quad (6)$$

Alternatively, the optimization approach uses regression with an objective function to minimize error

$$\min \left[\sum_{j=1}^N |y_{n,j} - \mathbf{R}_{P,n}| \right] \text{ subject to } 0 \leq p_{ii} \leq 1 \quad \text{for } i = 1, 2, \dots, 9 \quad (7)$$

In Eq. (7), N denotes the number of bridges belonging to a subset; $y_{n,j}$ is the observed condition rating at the n th stage of the j th bridge.

The optimization approach is significantly affected by maintenance actions that are difficult to code within current bridge management system inspection formats (Morcous 2006). Additionally, this approach is sensitive to the selection of an initial state vector.

Zoning Technique

Among the candidate variables, the *year built* has been identified as the most significant factor for the decrease in the deck element condition rating and is nearly always included for bridge management purposes anecdotally (Chang et al. 2017). The effect of bridge age

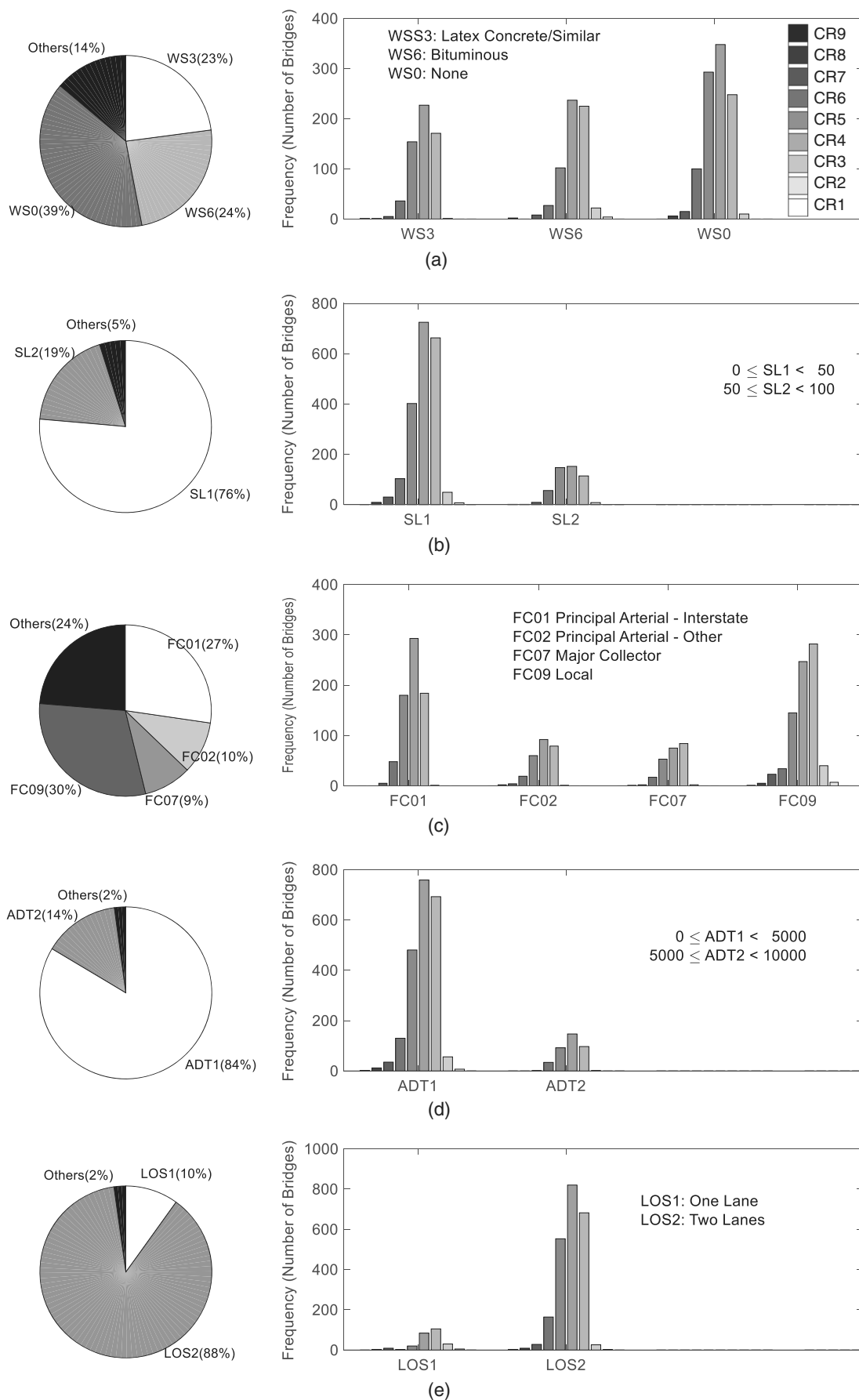


Fig. 1. Distribution of Wyoming bridges according to explanatory variables: (a) type of wearing surface; (b) structure length; (c) functional classification of inventory route; (d) average daily traffic; and (e) lanes on the structure.

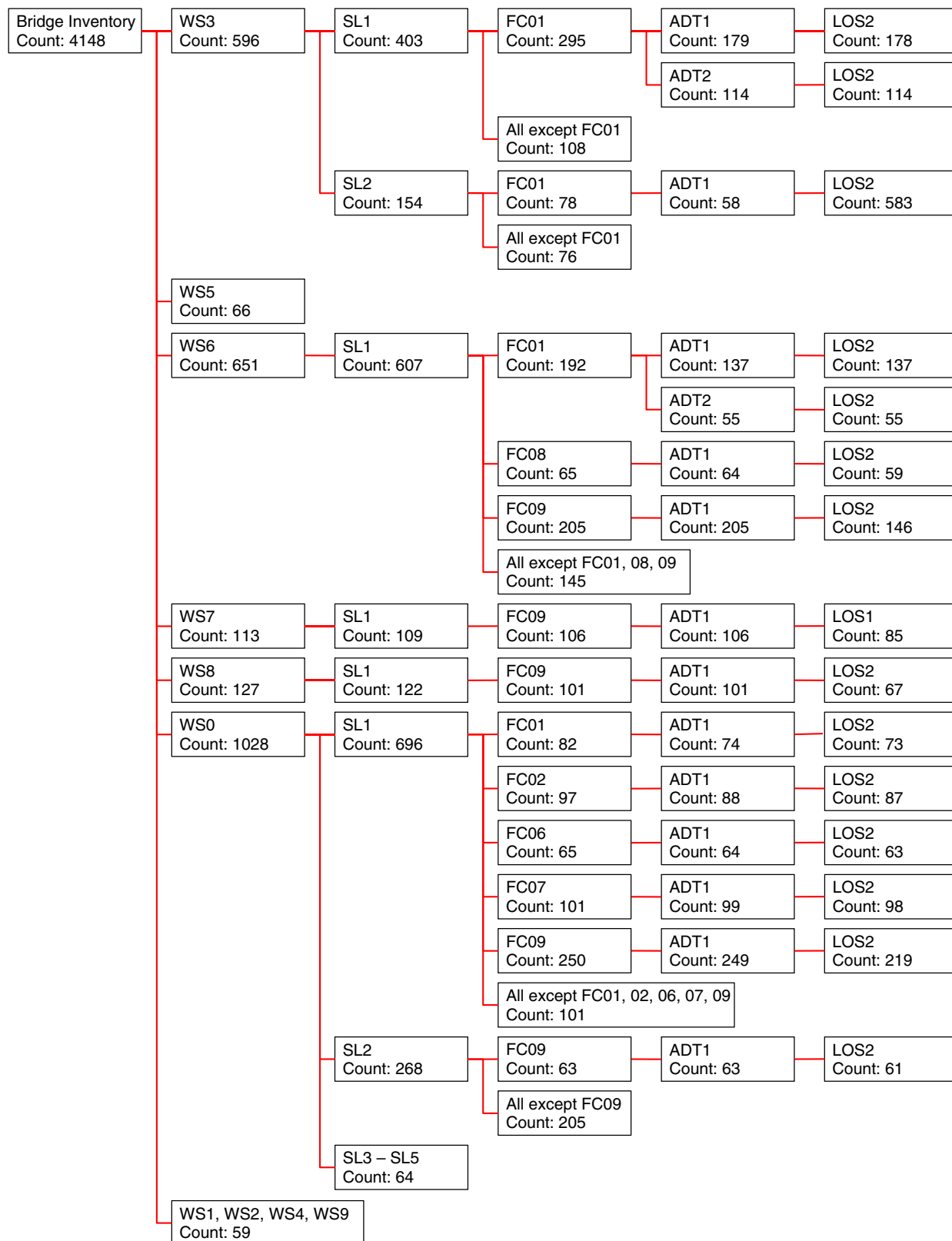


Fig. 2. Concept diagram for classification tree up to fifth level.

is often considered using a zoning technique (Butt et al. 1987; Jiang et al. 1988). In this paper, a 10-year interval is used to group bridge condition rating information, and the transition probability matrix is estimated for each zone. The initial condition rating vector for the $(i + 1)$ th zone is updated for every 10-year interval

$$\mathbf{q}_{0,j+1} = \mathbf{q}_{0,i} \mathbf{P}_i^{10} \quad (8)$$

An example of the changes of the probability matrix based on percentage prediction is investigated for multiple zones in Table 1, with the condition information in parentheses. There is a 0.77

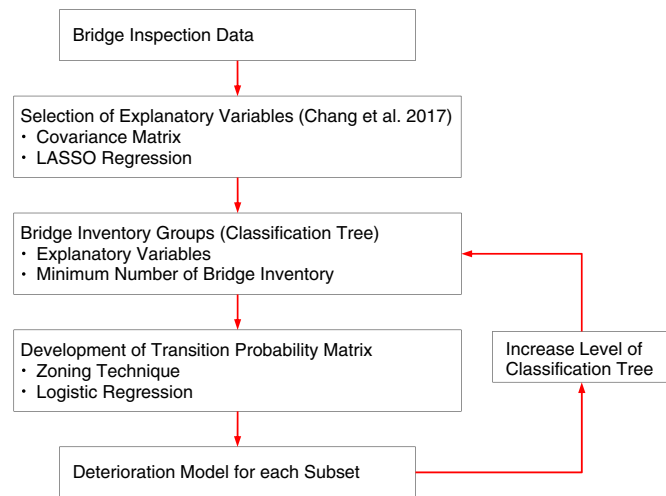


Fig. 3. Framework for the prediction update of condition rating.

Table 1. Diagonal elements of transition probability matrix for each zone

CR	Group 1	Group 3	Group 5	Group 7
9	0.77 (53)	1 (3)	0 (0)	0 (0)
8	0.78 (863)	0.77 (1,559)	0.70 (331)	0.82 (90)
7	0.92 (2,527)	0.89 (3,087)	0.90 (3,324)	0.89 (506)
6	0.96 (751)	0.94 (2,590)	0.94 (3,383)	0.94 (334)
5	0.97 (247)	0.96 (1,655)	0.94 (1,714)	0.95 (308)
4	0.95 (61)	0.93 (367)	0.95 (512)	0.91 (118)
3	1 (26)	0.97 (175)	0.96 (167)	1 (58)
2	0 (1)	0.95 (39)	0.99 (67)	1 (28)
1	1 (7)	1 (9)	1 (15)	1 (18)

probability that bridges less than a decade old (Group 1) are in CR9 and remain at this level. The probabilities for levels lower than CR4 are estimated using a relatively low number of bridge condition states. Group 7 indicates the large proportion of low condition rating structures so that the corresponding transition probability matrix is more suitable for relatively old bridges.

Logistic Regression

The components of a transition matrix indicate the probability of possible outcomes between adjacent inspection years. Typically, a transition matrix as shown in Eq. (3) specifies the conditional probability: whether the condition state of the bridge element will decrease or remain at the same stage compared to the current state. In the statistics literature, logistic regression is generally used to model the probability of binary or categorical outcomes that are dependent on predictor variables (Cox 1958). In the case of a binary or categorical outcome such as NBIS condition ratings, logistic regression is a statistically more appropriate tool for analysis. Specifically, this new approach models the logistic transformation of the probability as a linear function of the predictor variables.

Consider, for example, the case of a binary outcome, which refers to whether the condition rating of a bridge will deteriorate to the next stage or remain at the current stage. Without loss of generality, denote $\pi(\mathbf{X})$ as the probability of one outcome, depending on the explanatory variables $\mathbf{X} = [X_1, X_2, \dots, X_k]$. Then, the probability of the other outcome is $1 - \pi(\mathbf{X})$. The logistic transformation of $\pi(\mathbf{X})$ is defined below:

$$\log \left[\frac{\pi(\mathbf{X})}{1 - \pi(\mathbf{X})} \right] \quad (9)$$

This logistic transformation is also called *log odds* and it plays an important role in the analysis of binary data. Notice that although $\pi(\mathbf{X}) \in [0, 1]$, its log odds range from $-\infty$ to $+\infty$. This provides a sound mathematical basis for applying linear regression, as a linear function is unbounded. The logistic regression model is formally specified as

$$\log \left[\frac{\pi(\mathbf{X})}{1 - \pi(\mathbf{X})} \right] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k \quad (10)$$

In Eq. (10), β_j with $j = 0, \dots, k$ are regression coefficients. Alternatively, $\pi(\mathbf{X})$ is equivalent to the following:

$$\pi(\mathbf{X}) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)} \quad (11)$$

For this application, let $\pi(\mathbf{X})$ be the probability that an individual bridge will decrease in condition rating, and $\mathbf{X} = [X_1, X_2]$ with X_1 being the condition rating from the previous inspection year and X_2 the age. Then, $\pi(\mathbf{X})$ can be predicted by logistic regression as shown:

$$\pi(\mathbf{X}) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}{1 + \exp(\beta_0 + \dots + \beta_2 X_2)} \quad (12)$$

In Eq. (12), β_0 is the intercept and β_1 and β_2 are the coefficients corresponding to X_1 and X_2 , respectively. A set of bridges associated with the i th condition rating in previous years is grouped and each element of their probability vectors is averaged to represent the element of the transition probability matrix.

When focusing on a smaller data set, logistic regression is beneficial. The possibility of having a deficient condition is estimated with the logistic function so that smaller data sets can be used. For instance, a subset associated with [WS6, SL1, FC09, ADT1] was investigated. A traditional transition probability matrix can be estimated by the percentage prediction method (Table 2). The condition ratings from 7 to 9 are associated with a constant or decreasing stage, but bridges with condition ratings of 5 and 6 remain at the same stage, as shown by the values of 1 on the diagonal. This occurs because bridges under 10 years old primarily carry condition ratings of 7 and 8 (90%) and only 24 bridges (7.1%) received condition ratings under 6. The transition probability matrix that uses the optimization approach is presented in Table 3. This matrix minimizes the sum of errors between the predicted and the actual inspection using Eq. (7). The diagonal elements of Table 3 were estimated and the total probability theory was applied to calculate the off-diagonal elements. Since the actual inspection data do not include any components corresponding to condition ratings lower

Table 2. Transition probability matrix using counting number of bridges

CR	9	8	7	6	5	4	3	2	1
9	0.75	0.25	0	0	0	0	0	0	0
8	0	0.833	0.167	0	0	0	0	0	0
7	0	0	0.966	0.034	0	0	0	0	0
6	0	0	0	1	0	0	0	0	0
5	0	0	0	0	1	0	0	0	0
4	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1

Table 3. Transition probability matrix using optimization

CR	9	8	7	6	5	4	3	2	1
9	0.825	0.175	0	0	0	0	0	0	0
8	0	0.351	0.649	0	0	0	0	0	0
7	0	0	0.229	0.771	0	0	0	0	0
6	0	0	0	0.221	0.779	0	0	0	0
5	0	0	0	0	0.673	0.327	0	0	0
4	0	0	0	0	0	0.608	0.392	0	0
3	0	0	0	0	0	0	0.608	0.392	0
2	0	0	0	0	0	0	0	0.608	0.392
1	0	0	0	0	0	0	0	0	1

Table 4. Transition probability matrix using logistic regression

CR	9	8	7	6	5	4	3	2	1
9	0.734	0.266	0	0	0	0	0	0	0
8	0	0.856	0.144	0	0	0	0	0	0
7	0	0	0.967	0.033	0	0	0	0	0
6	0	0	0	0.993	0.007	0	0	0	0
5	0	0	0	0	0.999	0.001	0	0	0
4	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1

than 6, the solution for this case is not likely reliable and may provide a false sense of accuracy.

The logistic regression approach has the potential to resolve the issues caused by small sets of bridge data. The proportions used to formulate the transition probability matrix were modeled with a logistic regression model associated with the previous condition rating and the age. The numbers for each component were used as a weighting factor. Table 4 shows the transition probability matrix for the first group of the same subset, which can be used to successfully model a deficiency in the condition rating even though an insufficient number problem exists.

Discussion

Based on NBI reports from 1992 to 2015, Wyoming has operated 4,148 bridges over the past 24 years. The 2015 NBI report showed that 3,085 bridges were being inspected. Wyoming's total number of bridges is ranked 41st among states in the United States as of the 2013 bridge report. Wyoming is ranked third from the bottom, above Alaska and Nebraska, in number of bridges per unit area (Davis et al. 2013). Because Wyoming only has a small number of bridges, using a classification tree may produce subsets that are too small, limiting the accuracy of such a technique. Furthermore, simply using a single explanatory variable may also result in very small analysis sets. Additionally, to take advantage of the zoning technique, bridge data must be grouped using a homogeneous interval, which may further create issues with insufficient bridge numbers.

Deterioration Model Development without Classification Tree

Deterioration models were developed for all bridges using data from 1992 to 2005 and using an optimization-based approach, percentage prediction, and the logistic regression approach, as shown in Fig. 4. For each plot, three deterioration models corresponding to

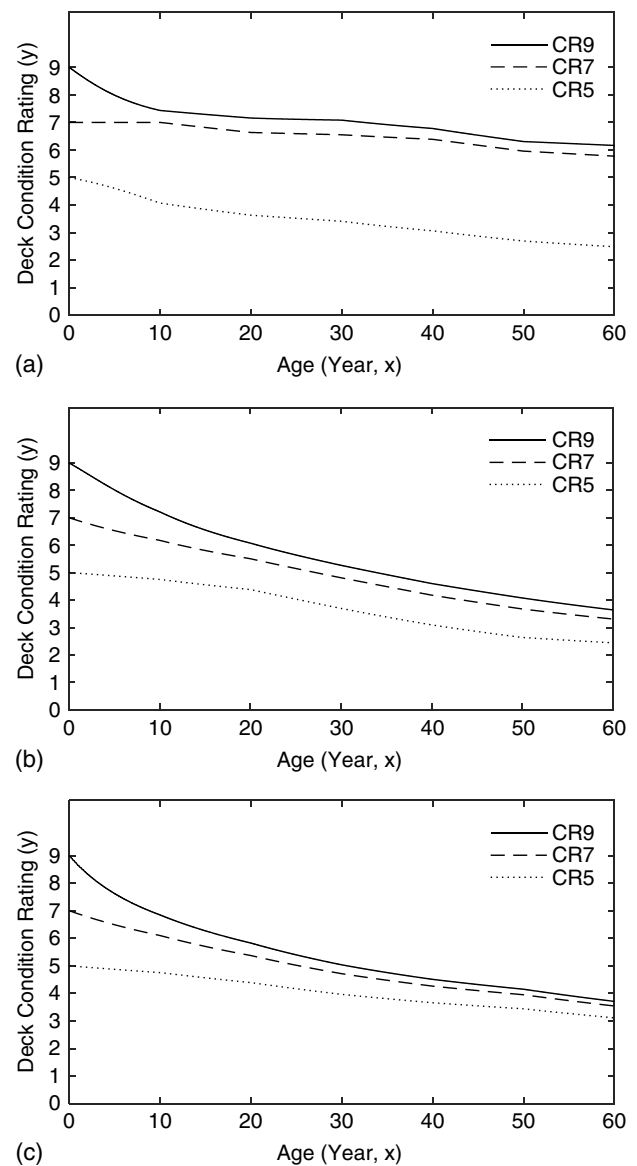


Fig. 4. Deterioration models for Wyoming bridges using: (a) optimization; (b) percentage prediction; and (c) logistic regression methods.

the condition ratings 5 (fair condition), 7 (good condition), and 9 (excellent condition) were developed. The initial state vector for the optimization-based approach was estimated using the distribution of bridges' condition ratings. For example, the distribution of new bridges' condition ratings was the first initial vector for the first zone and is defined as

$$\mathbf{q}_0 = [0 \quad 0.17 \quad 0.31 \quad 0.24 \quad 0.14 \quad 0.07 \quad 0.07 \quad 0 \quad 0] \quad (13)$$

The condition ratings for the newly constructed or replaced bridges are distributed mostly between a condition rating of 5 (fair condition) and 8 (very good condition). The initial vector was then replaced with a new distribution vector for the lowest age of every zone.

When using deterioration models that use optimization, the CR9 curve showed a fast decrease in performance, whereas the CR7 curve was almost constant at the beginning of the plot. The CR5 curve decreases quickly for the first decade and becomes almost linear after that time. The deterioration models for the

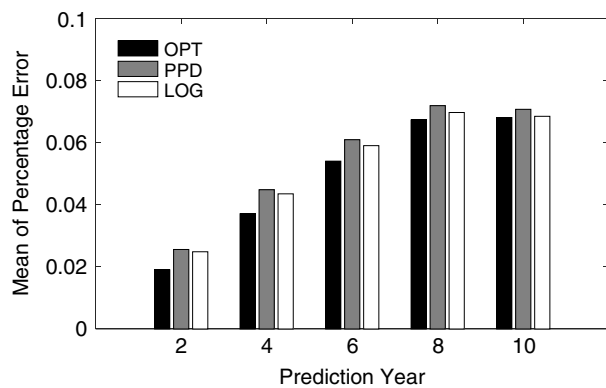


Fig. 5. Prediction error comparison between prediction and inspection for next 10 years using all data sets.

percentage prediction and logistic regression are similar. The CR9 curve for logistic regression deteriorates slightly faster, and the CR5 curve for logistic regression decreases less than the percentage prediction. The CR7 curve deteriorated in a nearly linear trend for both cases.

The prediction error was estimated by taking the mean absolute average error between the prediction and the inspection for the next 10 years, without breaking the data set into subsets using the classification tree, as shown in Fig. 5. For individual bridges, the elements of the initial state vector at 2004 and 2005 were defined as zeros except for the condition rating. The inspection records were tracked for bridges that were functioning in 2004 and 2005. The state vector $\hat{q}_{i,j}$ for the years' prediction, j , of the i th bridge with the age of n can be estimated as

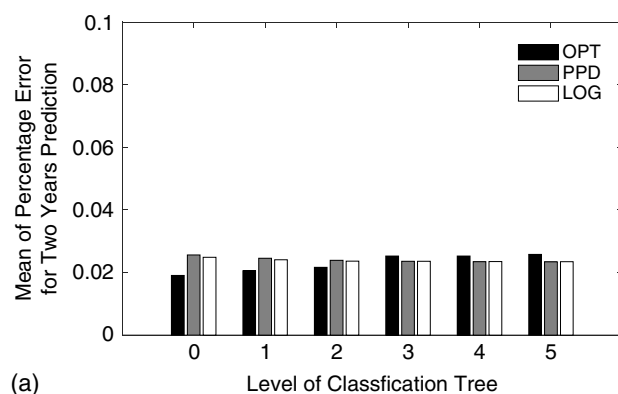
$$\hat{q}_{i,j} = \begin{cases} q_0 \cdot P_k^j & \text{if } j + n \leq 10 \cdot (k + 1) \\ q_0 \cdot P_k^{10-n} \cdot P_{k+1}^{j-10+n} & \text{otherwise} \end{cases} \quad (14)$$

In Eq. (14), P_k denotes the transition probability matrix for the k th age group. The condition rating is then predicted by taking the dot product of $\hat{q}_{i,j}$ and the regular condition vector $[9 \ 8 \ 7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1]^T$.

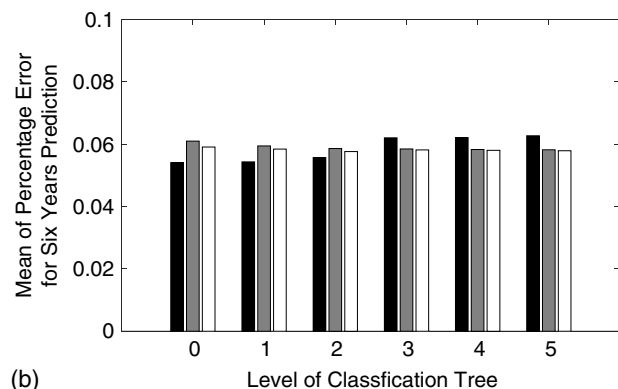
For the early prediction years, the smallest error is observed when optimization-based regression was used. The average error, however, increased relatively quickly compared to the other two methods so that it became worse than the logistic regression when the prediction year was larger than 10 years. This result was a significant factor for aiding the decision-making process when setting up an effective bridge management system, including budget assignment and a monitoring plan. In general, the elements of the transition probability matrix from an optimization approach are manipulated to minimize the sum of the error from the previous inspections. Therefore, the optimization-based approach fits very well for the near-future condition but is less reliable for the long-term prediction due to the increase of uncertainty in the transition probability matrix. The result of logistic regression is better than that of percentage prediction in general, since it can create nonzero probability for the transition probability matrix when only a small sample size is available.

Deterioration Model Development Considering Classification Tree

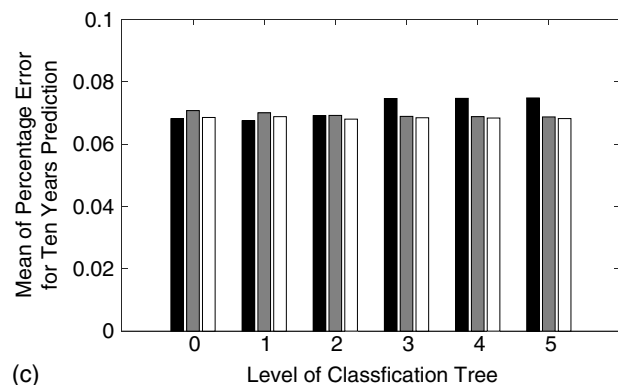
The effect of using a classification tree was investigated using the relative error comparison. The classification tree was used to split



(a)



(b)



(c)

Fig. 6. Prediction error comparison depending on the level of classification tree for (a) 2 years; (b) 6 years; and (c) 10 years' future inspection.

the bridge inspection data into five levels, which created 7, 8, 18, 15, and 15 individual deterioration models for the first through fifth levels, respectively. The prediction of the condition rating was updated with the increase in the classification tree level, and the error was evaluated for the optimization-based approach, percentage prediction, and logistic regression, as illustrated in Fig. 6. The relative average error was investigated to predict condition ratings 2, 6, and 10 years into the future.

The relative error increased as predictions were made further into the future, as expected. The relative error generally decreased for the percentage prediction and the logistic regression when the classification tree level increased. The small number of observations and the uncertainty caused by additional prediction years decreased the accuracy of the prediction. The logistic regression method was the most accurate for the 10-year prediction comparison. Similar to the comparison of all the data shown in Fig. 5, the relative error from logistic regression was better than that of the

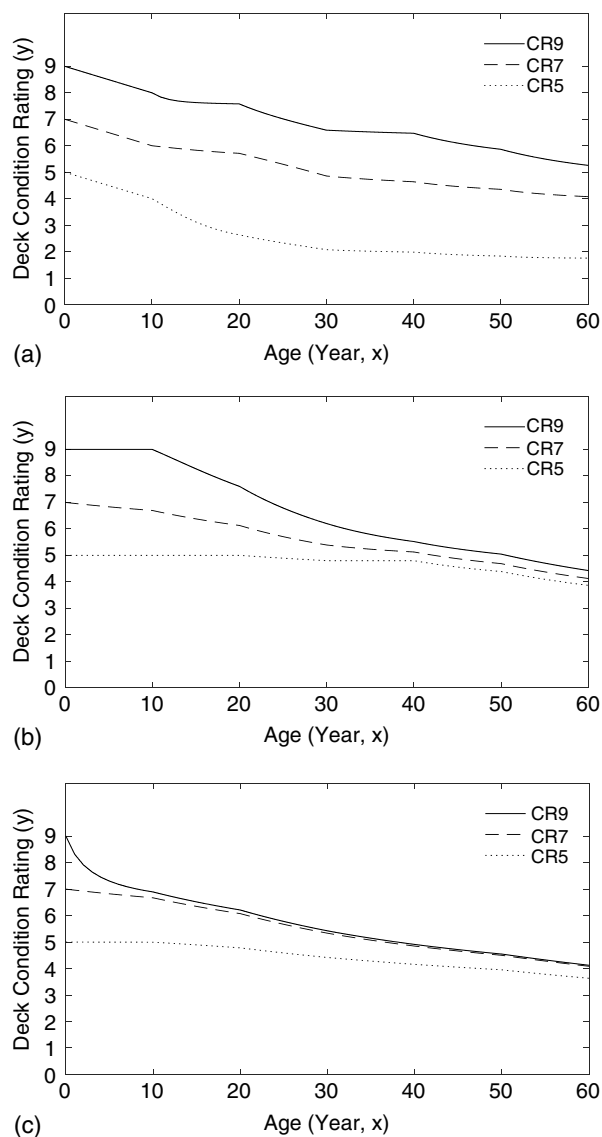


Fig. 7. Deterioration models for a subset of Wyoming bridges using: (a) optimization; (b) percentage prediction; and (c) logistic regression methods.

percentage prediction method. It does seem that the addition of the classification tree provides relatively little benefit when combined with the optimization approach; however, it does provide some benefit when combined with percentage prediction or the logistic regression method.

Deterioration Models for Small Data Set

The relative error comparison verified that logistic regression works well for decision makers who are seeking to establish a long-term monitoring plan. The uncertainty in the Markov chain is distributed well enough to generate a transition probability matrix with a small data set. The specific subsets are investigated to observe the effect of each method.

For example, consider a subset associated with WS0 = none; $0 < \text{SL1} < 50$ m; FC09 = rural (local); $0 < \text{ADT1} < 5,000$; and LOS = two lanes. This subset includes 1,731 samples from 219 bridges. The deterioration model was developed for this subset, as shown in Fig. 7. The relatively small number of samples in

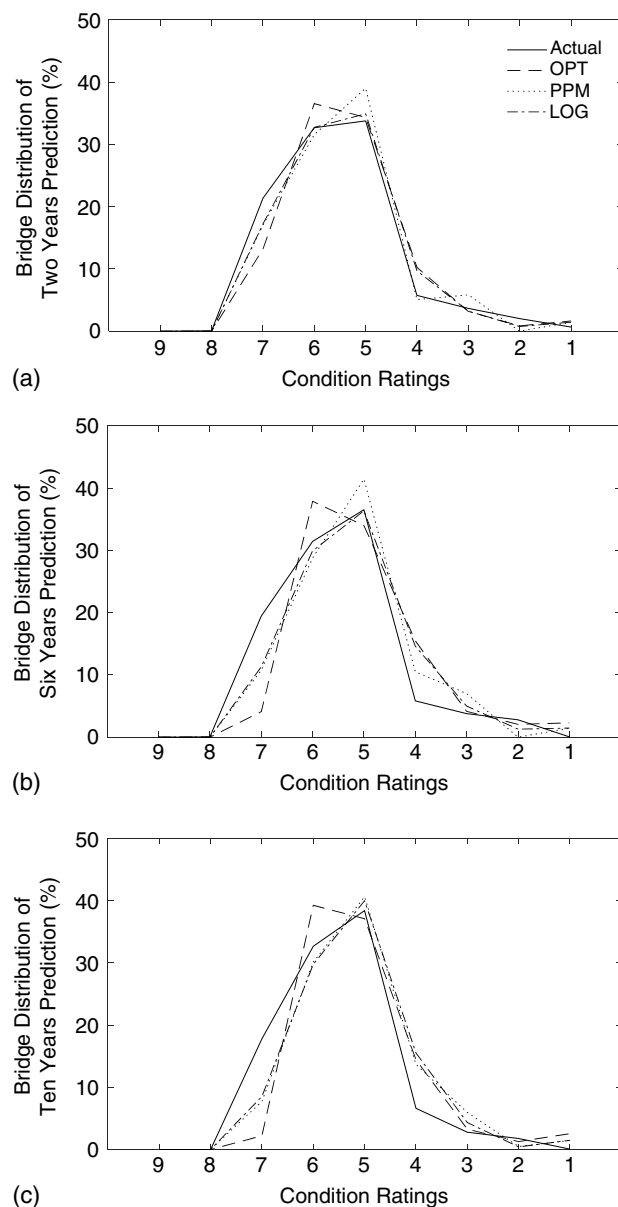


Fig. 8. Percentage distribution of bridges for a subset associated with WS0, SL1, FC09, ADT1, and LOS 2 when the prediction varies for (a) 2 years; (b) 6 years; and (c) 10 years.

this subset significantly affect the deterioration model for the optimization-based method, as shown in Fig. 7(a). The curve is not connected smoothly between adjacent age zones, and the decrease in the condition rating from 5 (fair) is rapid until it reaches a condition rating near 2 (critical). The percentage prediction is unable to model the decrease from the condition rating of 9 (excellent) for the first age zone since the first element of the transition matrix for this zone is unity, as shown in Fig. 7(b). Applying logistic regression shows a rapid drop at the early age of the deterioration model starting from a condition rating of 9 (excellent), whereas other findings are similar, as shown in Fig. 7(c).

The average distribution of the condition rating was estimated for each prediction year and plotted with the actual distribution (Fig. 8). The probability distribution for 2-year prediction shows that all investigated methods predict the future condition state fairly accurately. The percentage distribution shifts toward the right as the prediction year increases. Similarly, the distribution graph from all

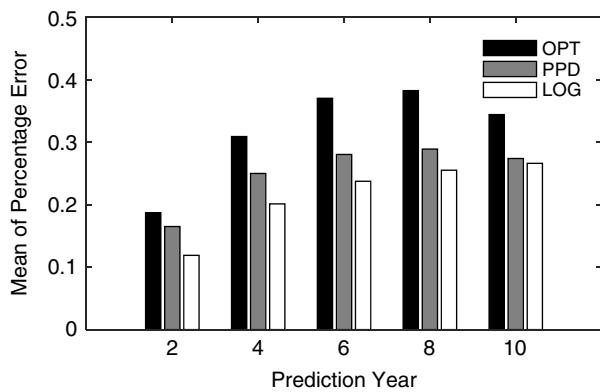


Fig. 9. Mean of percentage error in condition state vector versus prediction years.

the investigation methods moves toward the low condition ratings. Due to the mathematical procedure in the Markov chain, the proportion with the condition rating of 1 (imminent failure) increases over time whereas almost zero population is expected from the actual inspection data. The distribution error is estimated by taking the mean of percentage error between the actual distribution vector and the predicted vector from each method, as shown in Fig. 9. In general, the amount of error increases over prediction years and the logistic regression method shows better performance in the near term, but more so for the long-term predictions.

Conclusion

This paper presents a method to develop stochastic deterioration models using logistic regression as demonstrated on 24 years of Wyoming bridge inspection data. A Markov chain was employed to model uncertainty using a transition probability matrix with a classification tree and a zoning technique to investigate their accuracy and to create smaller subsets. Deterioration models for deck elements were developed and compared to the results from the percentage prediction, optimization, and logistic regression methods. From this comparison study, the following conclusions can be made:

1. The shapes of the deterioration models for logistic regression and percentage prediction are similar when a sufficient number of samples is used. The deterioration models are not smoothly connected nor continuous when relatively small data are available for the percentage prediction and the optimization methods. The logistic regression model seems less sensitive to issues with small data sets, indicating it may perform more accurately in these situations.
2. The classification tree can marginally improve prediction accuracies, but it decreases the size of subsets, leading to problems generating transition probability matrices. The use of these classification trees decreases the error for the logistic regression approach and the percentage prediction method but provides an increase in error for the optimization method, further indicating that the logistic regression method is better for small data sets.
3. Moving down the classification tree is only somewhat effective and provides little benefit past Level 2 (i.e., two explanatory variables). Additionally, the optimization method becomes less accurate when moving down the classification tree.
4. As expected, all methods become more inaccurate when predicting further into the future and are more accurate in near future predictions. However, it does seem that the error remains

somewhat constant after 8–10 years and even drops slightly for the optimization and percentage prediction methods.

5. The error comparison shows that the optimization method is best for predicting the condition rating in the near future; however, the uncertainty caused by the prediction year increases the error. Based on the data set investigated, the presented logistic regression provides a better long-term prediction.

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